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Prevention of Venous Thromboembolism in Internal Medicine with Unfractionated or Low-molecular-weight Heparins: A Meta-analysis of Randomised Clinical Trials

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Key words

Heparin, internal medicine, venous thromboembolism, prophylaxis, meta-analysis

Summary

Background. The prevention of venous thromboembolic disease is less studied in medical patients than in surgery. *Methods.* We performed a meta-analysis of randomised trials studying prophylactic unfractionated heparin (UFH) or low-molecular-weight heparin (LMWH) in internal medicine, excluding acute myocardial infarction or ischaemic stroke. Deep-vein thrombosis (DVT) systematically detected at the end of the treatment period, clinical pulmonary embolism (PE), death and major bleeding were recorded. *Results.* Seven trials comparing a prophylactic heparin treatment to a control (15,095 patients) were selected. A significant decrease in DVT and in clinical PE were observed with heparins as compared to control (risk reductions = 56% and 58% respectively, $p < 0.001$ in both cases), without significant difference in the incidence of major bleedings or deaths. Nine trials comparing LMWH to UFH (4,669 patients) were also included. No significant effect was observed on either DVT, clinical PE or mortality. However LMWH reduced by 52% the risk of major haemorrhage ($p = 0.049$). *Conclusions.* This meta-analysis, based on the pooling of data available for several heparins, shows that heparins are beneficial in the prevention of venous thromboembolism in internal medicine.

Introduction

It has been reported that 10% of the deaths observed in hospitals are related to pulmonary embolism and that 75% of these deaths occur in non-surgical patients (1-3). Some medical conditions, such as ischaemic stroke and heart failure, as well as some pre-existing patient-related variables (age, obesity, history of deep-vein thrombosis, etc.) have been clearly identified as high risk factors of venous thromboembolism (4). It seems moreover that more than 99% of the patients staying in hospitals present with at least one of these risk factors (5). These factors are moreover frequently combined: 80% of these patients present with at least three risk factors of thromboembolic disease (5).

The assessment of the prevention of venous thromboembolic disease is however less developed in non surgical patients than in surgery: nearly 100,000 patients have been included in trials concerning the prevention of venous thromboembolic disease in surgery (6-9) whereas only about 15,000 patients were included in such trials in medical care. The various meta-analyses carried out in surgery have moreover made it possible to quantify the risk of venous thromboembolic disease according to the type of surgery and to assess the benefits of the prophylactic antithrombotic treatments studied on venous thromboembolic disease (6-9). Such synthetic data were previously published for acute ischaemic stroke (10) and acute myocardial infarction (11) but no specific overview is available in the context of internal medicine.

We thus undertook to perform a meta-analysis of clinical trials performed in medical care, excluding cases of acute ischaemic stroke and acute myocardial infarction, in order to assess the prophylactic efficacy of heparins in such patients.

Materials and Methods

Study Selection

We performed a computer assisted literature search with Medline® and Current Contents® for randomised clinical trials comparing a heparin prophylaxis [with low-molecular-weight heparin (LMWH) or unfractionated heparin (UFH)] to a control in the prevention of venous thromboembolic disease in internal medicine. All the manufacturers of LMWH in France were contacted and asked to provide published or unpublished trial data relevant to internal medicine (Sanofi for Nadroparin, Rhône-Poulenc Rorer for Enoxaparin, Pharmacia-Upjohn for Dalteparin) so that the data used were exhaustive. Hand searching of reference lists of retrieved material was performed.

Only randomised trials assessing a dose of heparin corresponding to a prophylactic treatment of venous thromboembolic disease were included (i.e. for UFH: daily dose of 10,000 to 15,000 IU in two or three subcutaneous injections; for LMWH: doses corresponding to the prophylactic dose recommended for each one). Non randomised trials or trials assessing danaparoid sodium were excluded. The selection and extraction of data from the individual trials were initially and independently performed by two of the authors (PM, BT). In case of discrepancy between the results obtained by these two authors regarding either the selection of the studies or the extraction of data, a third author (SL) took the final decision.

In addition, a "best-study" subgroup was elaborated. This subgroup included only double-blind studies with a good randomisation method.

Evaluation Criteria

The primary end-point was the incidence of deep-vein thrombosis systematically detected at the end of the treatment period by isotopic fibrinogen uptake test, venous ultrasonography or venography of the lower limbs. The incidence of early mortality (from 10 days to 3 months), clinical pulmonary embolism confirmed by lung-scan or necropsy, and major haemorrhage were also studied.

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Statistical Considerations

The results of each trial were summarised on an intention-to-treat basis using 2 × 2 tables for each end-point in order to assess the antithrombotic efficacy of the different treatments. When the assessment of deep-vein thrombosis was not available, this criterion was considered as missing data, but the study was not excluded if another criterion was available (clinical PE, deaths or major bleeding). Since the results obtained from the different methods were similar (logarithm of the relative risk, logarithm of the odds-ratio, Mantel Haenszel, Cochran, and Peto), the meta-analyses were performed using the logarithm of the relative risk. A heterogeneity test was systematically performed and a random effect model for the relative risk was planned when this test was positive (12). Meta-analyses were performed using the software EasyMA® (13).

Results

Selected Studies

Eight trials assessing the prophylactic efficacy of heparin (LMWH or UFH) versus a control group of patients receiving no active antithrombotic treatment (with or without placebo) were found (14-21). One of them was excluded due to the system of randomisation applied (21): the allocation in this large open trial published by Halkin was indeed based on the regular hospital record number of each patient (odd or even), which made it possible to know before inclusion which treatment each patient would receive. In addition, eligibility of patients was determined only after randomisation and the proportions of eligible patients were significantly different between UFH and control. Since we were unable to exclude potential bias from this study, the study itself was excluded from the meta-analysis; it was however afterwards included in a sensitivity analysis. Seven randomised trials (3 trials with LMWH and 4 trials with UFH) were finally selected, which corresponds to a total of 15,095 patients (14-20). A summary of the study designs is given in Table 1. Four of these randomised studies were selected and included into a "best-study" subgroup. This subgroup excluded open studies (14, 17, 19) and/or without good randomisation method (19).

Nine clinical trials were found and selected for the comparison of LMWH and UFH (22-30), corresponding to a total of 4,669 patients (Table 2). Six of these nine randomised studies were included into a "best-study" subgroup which excluded three open studies (23, 25, 30).

Heparins versus Control

DVT was assessed in 5 trials (845 patients), early mortality in 4 trials (14,658 patients) and clinical PE and major haemorrhage in 6 trials (respectively 14,843 and 12,603 patients) (Table 3). The incidences of DVT and clinical PE in the absence of heparin therapy were respectively estimated at about 19% and 1%. The rate of mortality was of about 7%.

Taking into account the five trials systematically evaluating the incidence of DVT at the end of the treatment period, a significant 56% decrease in DVT was observed with heparins (relative risk = 0.44 [95% confidence interval 0.29 to 0.64]; P <0.001, Fig. 1). In addition, a significant reduction in clinical PE (relative risk = 0.48 [0.34 to 0.68]; P <0.001, Fig. 1) and a non-significant increase in the incidence of major haemorrhage (relative risk = 1.87 [0.94 to 3.75]; P = 0.08, Fig. 1) were observed with heparin but no effect on mortality was shown (relative risk = 0.95, Fig. 1). The heterogeneity test was non-significant for each comparison.

The study published by Halkin (21), which was first excluded due to its system of randomisation, was finally included in this meta-analysis to test the robustness of the results. Only deaths (at one month) were recorded in this study (52/669 deaths in the UFH group and 75/689 deaths in the control group). A non-significant reduction in mortality was also observed when this study was taken into account (relative risk = 0.92 [0.82 to 1.03]; P = 0.14).

The robustness of the results was checked thanks to the "best study" subgroup, which only includes double blind trials with an adequate randomisation method. The results were comparable in this subgroup

Table 1 Summarised designs of all randomised controlled trials of heparin therapy versus control

Authors, *year*	Type and dose of heparin (IU/day)	Treatment duration	Patients	No. of patients	Double blind	Assessment of DVT	Mortality	« Best-study » subgroup
Belch et al. *1981* (14)	UFH 15000	9 days	pulmonary infectious and heart failure	100	no	fibrinogen uptake test	ne	no
Cade et al. *1982* (15)	UFH 10000	5-9 days	critically ill and various	234	yes	fibrinogen uptake test	ne	yes
Dahan et al. *1986* (16)	LMWH : Enoxaparin 6300	10 days	various and elderly	270	yes	fibrinogen uptake test	10 days	yes
Ibara-Perez et al. *1988* (17)	UFH 10000	until discharge	various	85	no	fibrinogen uptake test	ne	no
Bergmann et al. *1996* (18)	LMWH : Nadroparin 3075	21 days	various	2474	yes	fibrinogen uptake test #	21 days	yes
Gärdlung et al. *1996* (19)	UFH 10000	8 days	infectious disease	11693	no	fibrinogen uptake test #	60 days	no
Fraisse et al. *1998* (20)	LMWH : Nadroparin 4100-6150*	until discharge	chronic obstructive pulmonary disease	239	yes	venography	21 days	yes

LMWH : low molecular weight heparin ; UFH : unfractionated heparin ; IU : international unit ; DVT : deep-vein thrombosis; ne : not evaluated.
The « best-study » subgroup excluded trials without double blind design (14,17,19) and/or poor method of randomisation (19).
* adjusted to body weight (45-70 kg : 4100, 71-110 kg :6150)
# not systematically performed at the end of the treatment period but used in case of clinical suspicion.

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Table 2 Summarised designs of all randomised controlled trials of LMWH versus UFH therapy

Authors, *years*	HBPM vs. UFH and dose (IU/day)	Treatment duration	Patients	No. of patients	Double blind	Assessment of DVT	Mortality	« Best-study » subgroup
Poniewierski et al. *1988* (22)	Dalteparin 2500 vs UFH 10000	7-10 days	various (low and high risk)	200	yes	ne	10 days	yes
Aquino et al. *1990* (23)	Nadroparin 3075 vs UFH 10000- 15000	10 days	various and elderly	99	no	ultra-sonography	10 days	no
Harenberg et al. *1990* (24)	Dalteparin 2500 vs UFH 15000	10 days	various	166	yes	IPG and ultra-sonography	10 days	yes
Forette et al. *1995* (25)	Nadroparin 3075 vs UFH 10000- 15000*	28 days	various and elderly	295	no	ultra-sonography	28 days	no
HESIM *1996* (26)	Nadroparin 3100 vs UFH 15000	10 days	various	1590	yes	ultra-sonography	10 days	yes
APTE *1996* (27)	Nadroparin 3075 vs UFH 10 000	90 days	various and elderly	256	yes	IPG and ultra-sonography	90 days	yes
PRIME *1996* (28)	Enoxaparin 4000 vs UFH 15000	7 days	various	959	yes	ultra-sonography	not specified	yes
EMSG *1996* (29)	Enoxaparin 2100 vs UFH 10000	10 days	various and elderly	439	yes	fibrinogen uptake test	10 days	yes
PRINCE *1998* (30)	Enoxaparin 4200 vs UFH 15000	10 days	cardio-pulmonary disease	665	no	venography	10 days	no

LMWH : low molecular weight heparin ; UFH : unfractionated heparin ; IU : international unit ; DVT : deep-vein thrombosis; ne : not evaluated ; IPG : Impedance phlethysmography.
The « best-study » subgroup excluded trials without double blind design (22,25,30) and/or poor method of randomisation (no study concerned).
* adjusted to body weight (40-70 kg : 10000, > 70 kg : 15000)

Table 3 Summarised results of all randomised controlled trials of heparin therapy versus control

	Deep-vein thrombosis (n/N)		Pulmonary embolism (n / N)		Death (n / N)		Major bleeding (n / N)	
Study	Heparin	Control	Heparin	Control	Heparin	Control	Heparin	Control
Belch (14)	2/50	13/50	0/50	2/50	ne	ne	0/50	0/50
Cade (15)	15/154	15/80	ne	ne	ne	ne	0/154	0/80
Dahan (16)	4/132	12/131	0/135	3/135	1/135	6/135	1/135	3/135
Ibara-Perez (17)	1/33	12/46	0/39	3/46	ne	ne	4/39	0/46
Bergmann (18)	ne	ne	10/1230	17/1244	124/1230	128/1244	ne	ne
Gärdlung (19)	ne	ne	34/5776	74/5917	304/5776	333/5917	14/5776	6/5917
Fraisse (20)	13/84	24/85	0/108	0/113	18/108	17/113	6/108	3/113
	35/453	76/392	44/7338	99/7505	447/7249	484/7409	25/6262	12/6341

n/N : number of events / number of patients ; ne : not evaluated.

Table 4 Summarised results of all randomised controlled trials of LMWH versus UFH

	Deep-vein thrombosis (n/N)		Pulmonary embolism (n / N)		Death (n / N)		Major bleeding (n / N)	
Study	LMWH	UFH	LMWH	UFH	LMWH	UFH	LMWH	UFH
Poniewierski (22)	ne	ne	0/100	0/100	1/100	0/100	ne	ne
Aquino (23)	1/49	1/50	0/49	0/50	1/49	2/50	0/49	2/50
Harenberg (24)	3/84	4/82	ne	ne	3/84	1/82	0/84	1/82
Forette (25)	3/146	3/149	0/146	1/149	6/146	7/149	0/146	4/149
HESIM (26)	6/726	4/710	5/810	6/780	23/810	9/780	5/810	4/780
APTE (27)	0/129	1/127	ne	ne	8/129	8/127	1/129	4/127
PRIME (28)	1/477	4/482	0/477	4/482	7/477	11/482	2/477	9/482
EMSG (29)	9/207	10/216	1/216	0/223	7/216	8/223	1/216	2/223
PRINCE (30)	19/239	22/212	1/239	1/212	28/332	30/333	1/332	1/333
	42/2057	49/2028	7/2037	12/1996	84/2343	76/2326	10/2243	27/2226

LMWH : low molecular weight heparin ; UFH : unfractionated heparin ; n/N : number of events / number of patients ; ne : not evaluated.

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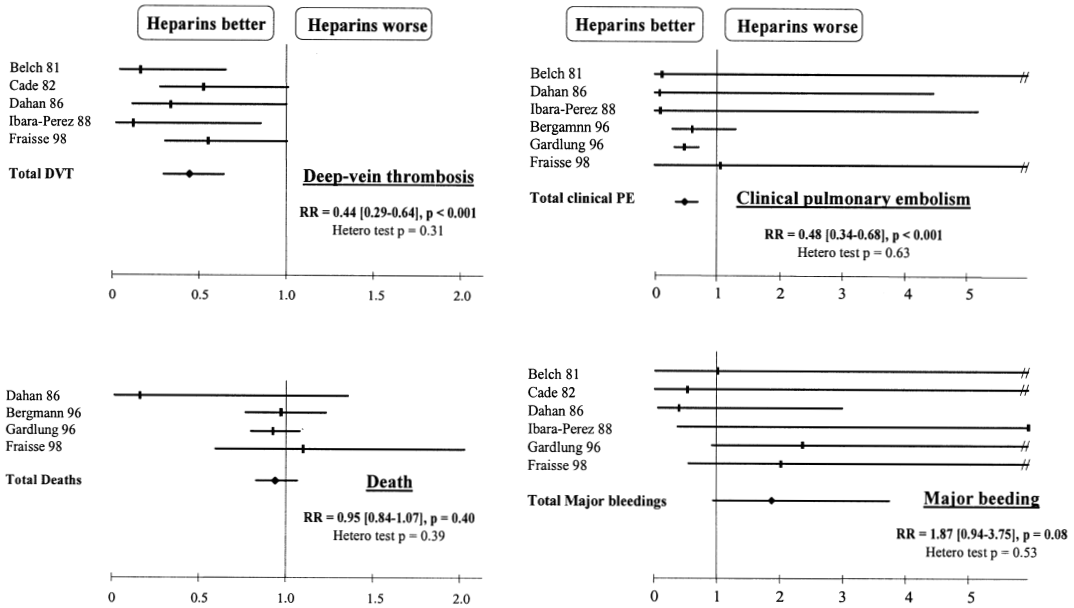

Fig. 1 Graphical representation of the results of the meta-analysis (logarithm of the relative risk method) for deep-vein thrombosis (upper left), pulmonary embolism (upper right), deaths (lower left) and major bleeding (lower right) when comparing heparin (UFH and LMWH) to control. A relative risk <1 indicates that heparins are more efficient than control. The horizontal lines represent the 95% confidence interval; if value of 1 is included the results are not significant. (//) denotes a value greater than 6 for the upper limit of the confidence interval

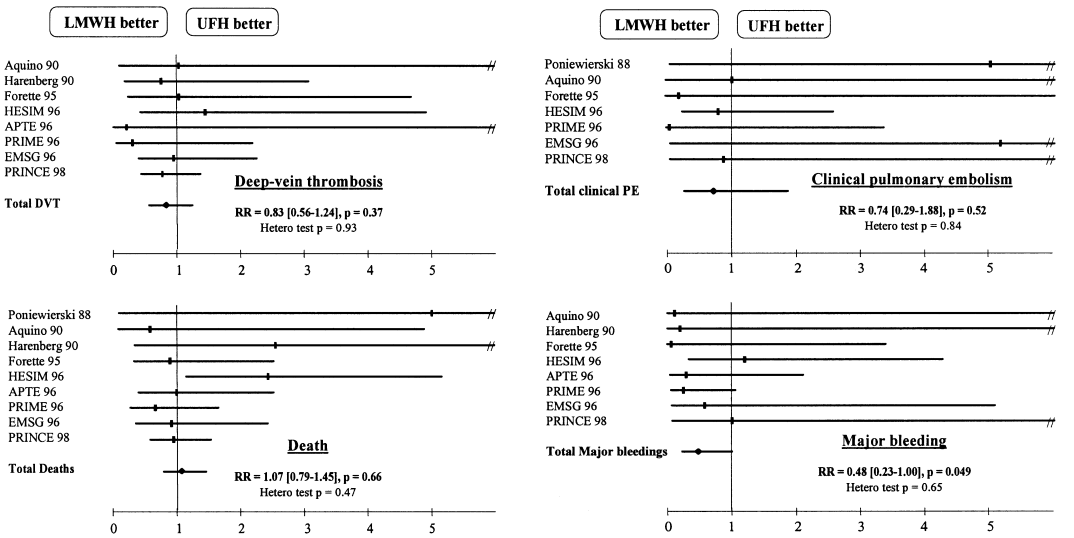

Fig. 2 Graphical representation of the results of the meta-analysis (logarithm of the relative risk method) for deep-vein thrombosis (upper left), pulmonary embolism (upper right), deaths (lower left) and major bleeding (lower right) when comparing LMWH to UFH. A relative risk <1 indicates that LMWH is more efficient than UFH. The horizontal lines represent the 95% confidence interval; if value of 1 is included the results are not significant. (//) denotes a value greater than 6 for the upper limit of the confidence interval

despite the loss of power, with a relative risk of 0.50 [0.33 to 0.76] for DVT (666 patients); 0.57 [0.27 to 1.20] for clinical PE (2,965 patients); 0.98 [0.79 to 1.21] for total number of deaths (2,965 patients); and 1.20 [0.41 to 3.56] for major bleeding episodes (725 patients).

LMWH versus UFH

DVT and major haemorrhage were assessed in 8 trials, with respective totals of 4,085 and 4,469 patients (DVT assessment was not available for all patients). Early mortality was studied in all trials (4,669 patients) and clinical PE in 7 trials (4,033 patients) (Table 4).

No significant difference between LMWH and UFH was observed concerning the incidence of DVT (relative risk = 0.83 [0.56 to 1.24]; P = 0.37), the incidence of clinical PE (relative risk = 0.74 [0.29 to 1.88]; P = 0.52) and the mortality (relative risk = 1.07 [0.79 to 1.45]; P = 0.66, Fig. 2). However, LMWH reduced the risk of major haemorrhage in comparison with UFH (relative risk = 0.48 [0.23 to 1.00], P = 0.049, Fig. 2). The heterogeneity test was non-significant for each comparison.

The results were comparable based on the "best study" subgroup, with a relative risk of 0.87 [0.49 to 1.59] for DVT (3,240 patients); 0.76

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[0.27 to 2.19] for clinical PE (3,188 patients); 1.26 [0.82 to 1.93] for total number of deaths (3,610 patients) and 0.51 [0.23 to 1.12] for major bleeding episodes (3,410 patients).

Discussion

Our results show that the risk for DVT of the lower limbs is of about 19% in internal medicine when no prophylactic treatment is administered. In the absence of heparin therapy, the risk for DVT in internal medicine is comparable to that observed in general surgery (6-7, 9) and in cases of acute myocardial infarction (11) but lower than that observed in cases of acute ischaemic stroke (10) or in orthopaedic surgery (6-9). It was in fact very difficult to assess the real risk of DVT in internal medicine due to the different methods used to assess DVT in the clinical trials we selected, i.e. the different diagnostic methods, for example fibrinogen uptake test, ultrasonography and venography of the lower limbs.

Despite the heterogeneity of the methods used for the diagnostic of DVT, there is an homogeneous treatment to control ratio whatever the diagnostic method applied was (non significant heterogeneity test). This meta-analysis shows that heparins involve a 56% risk reduction of DVT of the lower limbs. This prophylactic effect of heparin in DVT is similar to that observed in other medical [acute myocardial infarction (11) or acute ischaemic stroke (10)] or surgical situations (6-9), with a risk reduction of DVT assessed at about 60%.

The clinical relevance of these DVT systematically detected at the end of the treatment period is questionable since most of these were asymptomatic and distal. We however also observed a significant 52% risk reduction in clinical and fatal PE in internal medicine (Fig. 1). Even if heparin appeared to have no effect on mortality, the clinical benefit on PE associated to a small incidence of major bleedings yields a highly favourable benefit-risk ratio for heparin therapy in the prevention of venous thrombo-embolism in internal medicine.

This recommendation is however not definitive, since the patients of the different studies pooled were enrolled in internal medicine studies based on a confinement to bed for a period of more than 72 h which thus includes heterogeneous pathologies in terms of thrombotic risk (4, 31). This is why even if the mean deep-vein thrombosis risk observed for the category "internal medicine" amounts to 19%, some of the pathologies concerned may be associated with a too low thrombotic risk to require a prophylactic treatment. A prophylactic heparin treatment thus remains to be assessed in this indication in order to estimate the real benefit-risk ratio for different sub-populations stratified according to their DVT risk.

Our meta-analysis shows that LMWH and UFH have a similar efficiency but that LMWH are associated with a significantly lower risk of major bleedings (relative risk = 0.48, p = 0.049). Despite the different definitions of major bleeding among the trials (three studies with major bleeding defined as overt haemorrhage involving cessation of treatment, three studies with major bleeding defined as intracranial or retroperitoneal haemorrhage or as haemorrhage requiring transfusion and two studies without clear definition), the relative risks seem to be constant across the studies according to the heterogeneity test (p = 0.59).

Concerning efficacy, the power of our meta-analysis is too small to detect a difference between LMWH and UFH. However, according to their better benefit-risk ratio observed in surgery (7, 9) and due to their convenience in comparison with UFH, the validation of a prophylactic heparin treatment in internal medicine might be assessed using LMWH rather than UFH.

This meta-analysis seems to validate the necessity of a heparin therapy as a prevention against venous thrombo-embolism in internal medicine. This prophylactic indication can however not be generalised without first taking into account the level of thrombotic risk to which the patient is exposed; the number of major bleeding episodes is indeed higher with heparin than in the absence of treatment. It thus remains necessary to determine up to which level of thrombotic risk the benefit-risk ratio is still in favour of the administration of heparin.

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